Laboratory 1: Viscous Gravity Current

ENGN 0810

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Introduction

The purpose of this lab is to investigate the spreading behavior of viscous fluids under the force of gravity, otherwise known as viscous gravity currents. Examples of this phenomenon include lava flows, avalanches, and the spread of pollutants. The spreading process is the result of a balance between gravity and viscosity.

The basis of this experiment was Huppert's (1982) prediction of the radius of the front current of viscous-dominated flows. Assuming a Reynolds number of $Re < 1$ and a Bond number of $Bo > 1$, the radius of the spreading front is

$$R(t) = 0.779\left(\frac{gpV^3t}{\mu}\right)^{1/8}$$

The goal of this lab was to measure the viscosities of two viscous fluids by pouring them on a flat surface, allowing us to compare experimental results with theoretical calculations.

Methods

The experiment was conducted using two viscous fluids on a flat horizontal surface. Each fluid was poured three times using three different volumes, at the center of the surface. Using a smartphone camera mounted to a tripod, the flowing fluid was video recorded until negotiable growth was shown. To calibrate measurements, a ruler was placed on the board at the beginning of every trial.

Each fluid was poured on to the surface at 30, 40, and 50 mL. The fluid samples were poured quickly and consistently to reduce variability between trials.

Video data was analyzed using MATLAB. For each trial, the radius of the spreading fluid was measured at approximately 20 time intervals, using pixel count to then calculate physical distance. These measurements were then used to compare the fluid's physical spreading with theoretical predictions, ultimately estimating their viscosity.

Results

The figures below show the velocity vs. time graphs for the six measured volumes. The blue dots represent the experimental measurements and the red line represents the theoretical velocity vs. time graph:

Figure 1: Yellow Solution (30ml pour) [Velocity vs. Time]

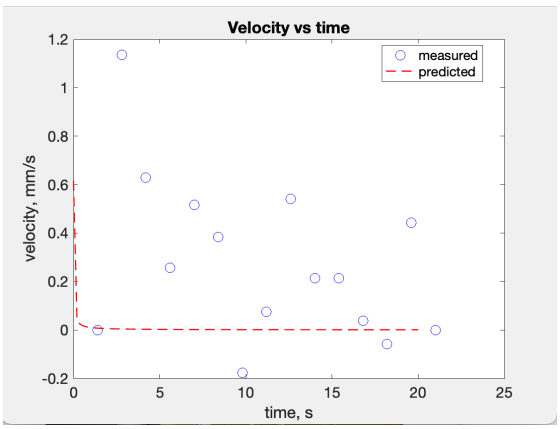


Figure 2: Yellow Solution (40 ml pour) [Velocity vs. Time]

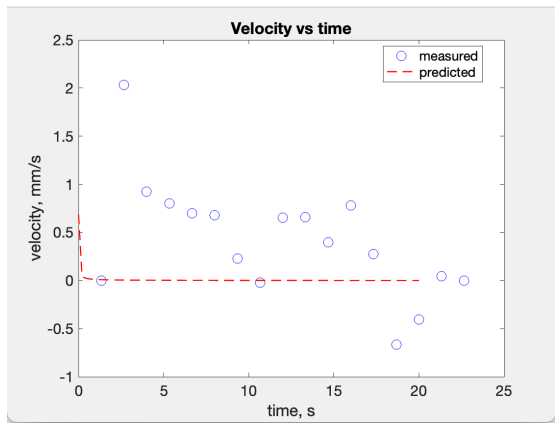


Figure 3: Yellow Solution (50ml pour) [Velocity vs. Time]

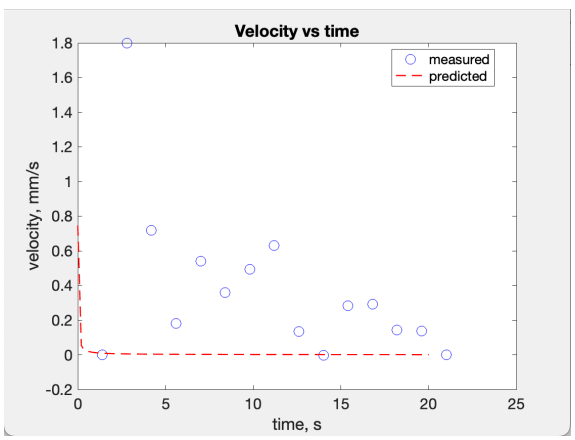


Figure 4: Blue Solution (30 ml pour) [Velocity vs. Time]

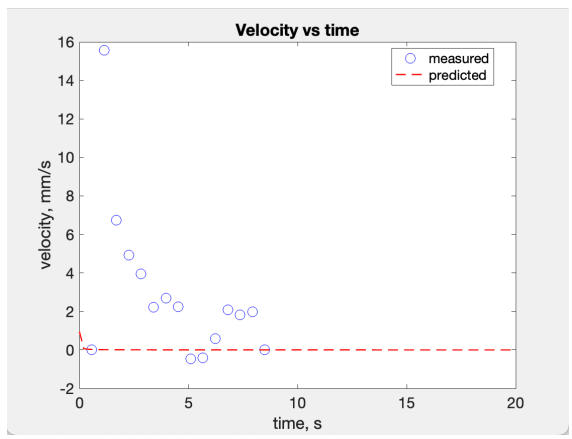


Figure 5: Blue Solution (40ml pour) [Velocity vs. Time]

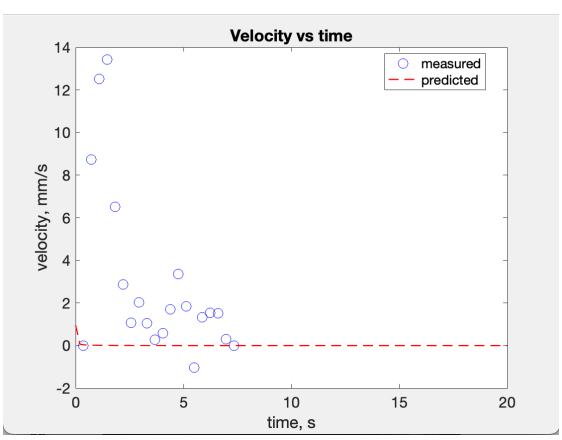
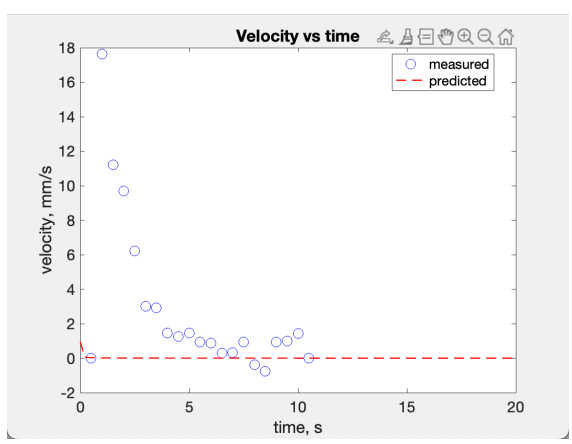


Figure 6: Blue Solution (50 ml pour) [Velocity vs. Time]



The figures below show the diameter vs time graphs for the six measured volumes. The blue dots represent the experimental measurements and the red line represents the theoretical diameter vs time graph.

Figure 7: Yellow Solution (30ml pour) [Diameter vs. Time]

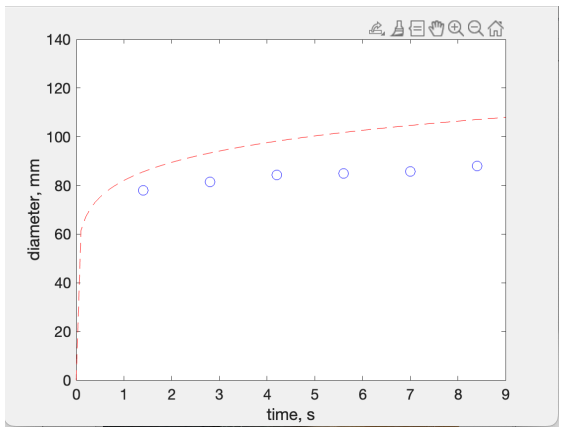


Figure 8: Yellow Solution (40 ml pour) [Diameter vs. Time]

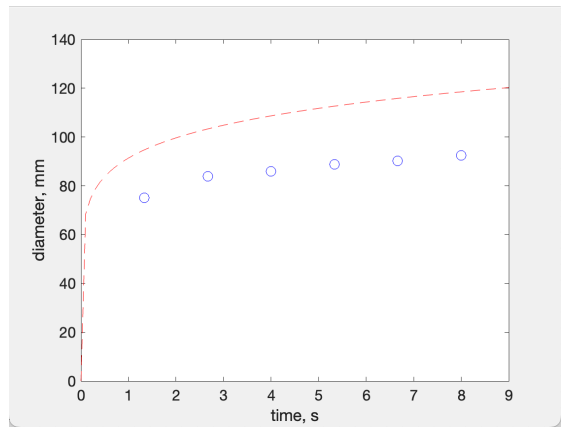


Figure 9: Yellow Solution (50ml pour) [Diameter vs. Time]

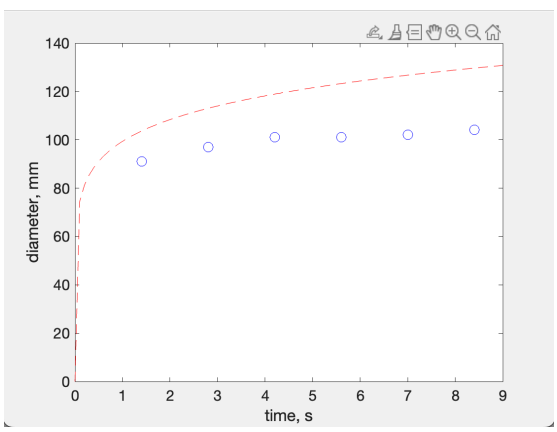


Figure 10: Blue Solution (30 ml pour) [Diameter vs. Time]

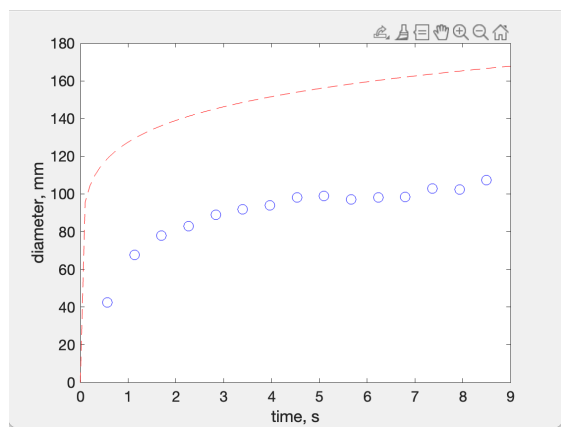


Figure 11: Blue Solution (40 ml pour) [Diameter vs. Time]

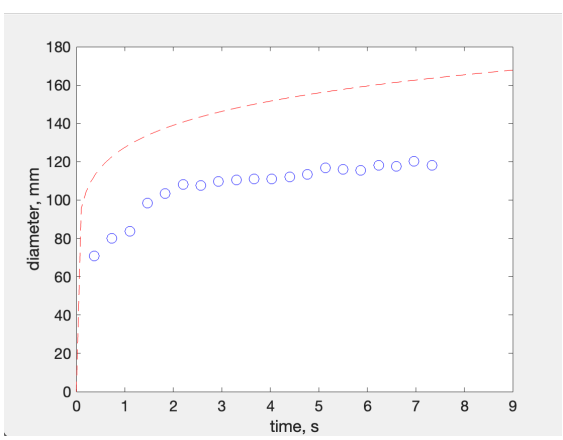
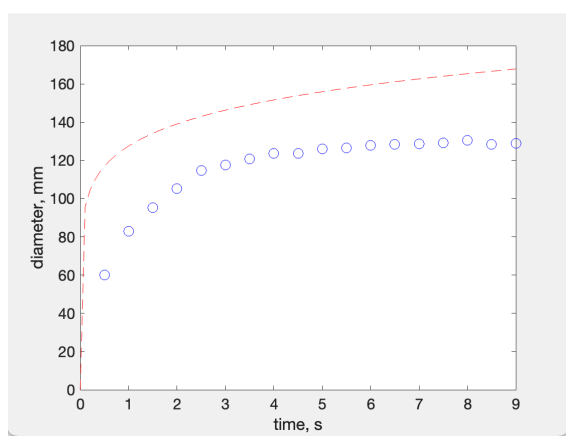


Figure 12: Blue Solution (50 ml pour) [Diameter vs. Time]



The figures below show the diameter vs time graphs on a logarithmic scale for the six measured volumes. The blue dots represent the experimental measurements and the black line represents the theoretical diameter vs. time graph.

Figure 13: Yellow Solution (30ml pour) [Diameter vs. Time]

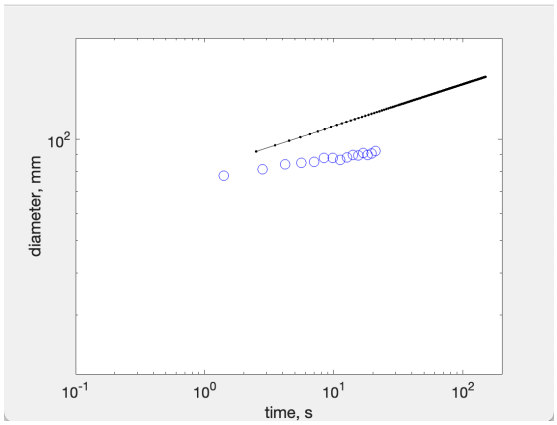


Figure 14: Yellow Solution (40 ml pour) [Diameter vs. Time]

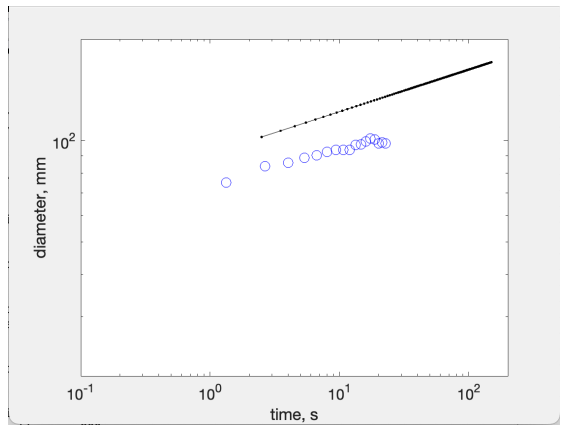


Figure 15: Yellow Solution (50ml pour) [Diameter vs. Time]

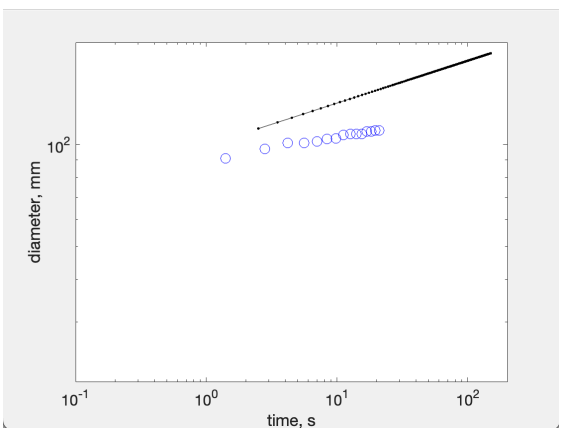


Figure 16: Blue Solution (30 ml pour) [Diameter vs. Time]

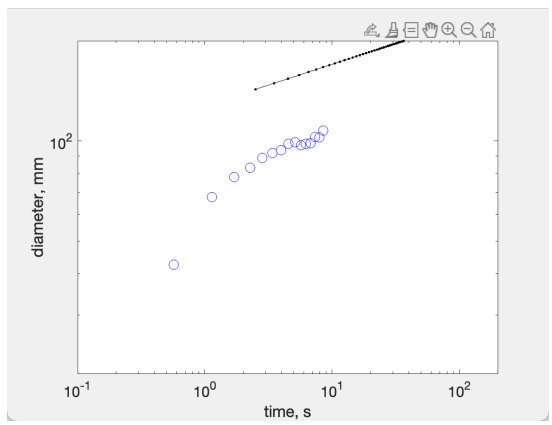


Figure 17: Blue Solution (40 ml pour) [Diameter vs. Time]

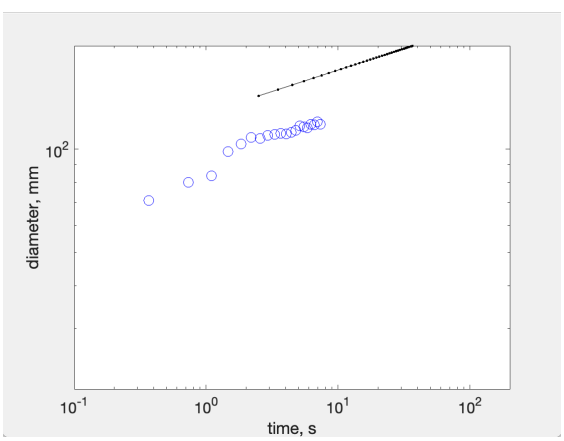
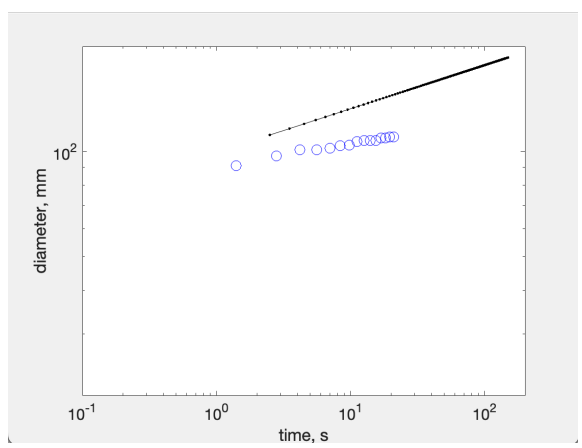


Figure 18: Blue Solution (50 ml pour) [Diameter vs. Time]



The figures below show the diameter vs time graphs where the radius is replaced by the dimensionless variable R* and time is replaced by the dimensionless variable t*.

Figure 19: Yellow Solution (30ml pour) [R vs. T*]*

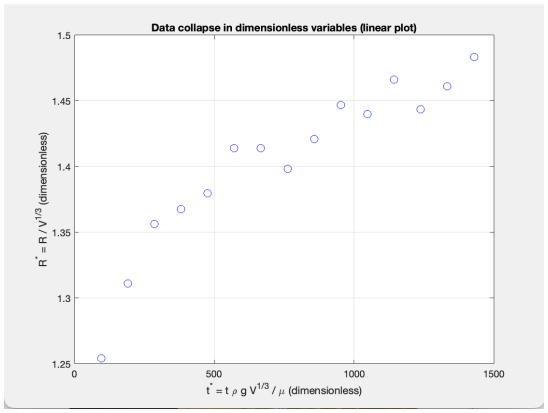


Figure 20: Yellow Solution (40 ml pour) [R vs. T*]*

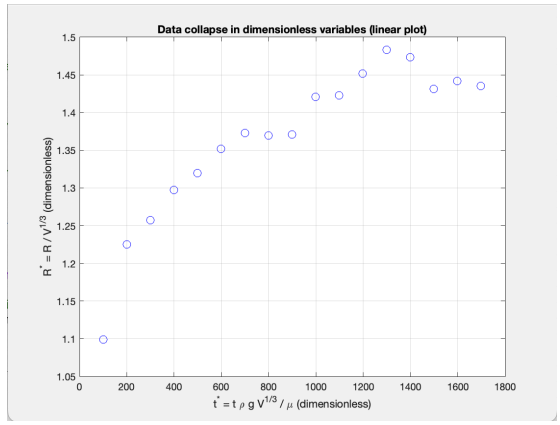


Figure 21: Yellow Solution (50ml pour) [R vs. T*]*

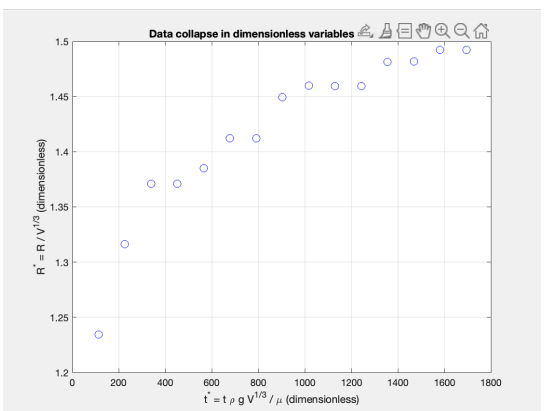


Figure 22: Blue Solution (30 ml pour) [R vs. T*]*

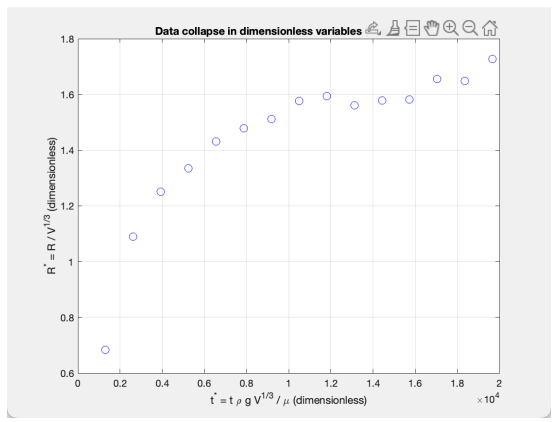


Figure 23: Blue Solution (40ml pour) [R vs. T*]*

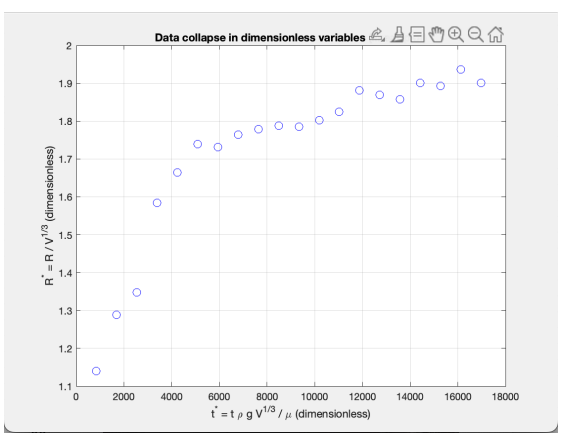
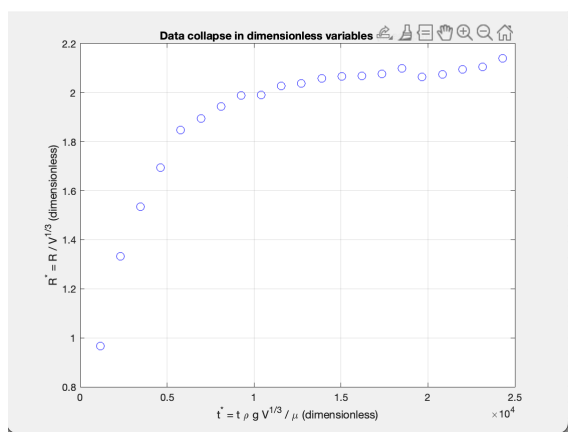


Figure 24: Blue Solution (50 ml pour) [R vs. T*]*



Discussion, Conclusions, and Recommendations

(2) Figures 1-6 show the calculated velocities of the liquid samples with respect to time. Our velocity vs time graphs were all riddled with outliers and the data was not consistent with what we predicted theoretically. This can be mostly attributed to error when clicking on the screen to measure the diameters of the fluids, where being a few pixels off could result in a drastic change in the calculated velocity. The plots were made using the following equation applied to the diameter vs time plots:

$U^i \approx \frac{R^{(i+1)} - R^{(i-1)}}{t^{(i+1)} - t^{(i-1)}}$ *[U = velocity, R = diameter, t = time, i = data point index]*

Nevertheless, the pattern that can be observed is one where an increase in fluid volume leads to an increase in the initial velocity of the gravity current. The velocities of the blue liquid were also much higher than those of the yellow liquid, due to the yellow liquid having a much greater viscosity.

(3) Figures 7-12 show the radius vs time graphs. Our radius vs time graphs were far more consistent with the theoretical lines, however at every point in time, the experimental radius was under the theoretical. This can be attributed to multiple causes including error in measuring the radii and a slight incline on the table over which we poured the fluids. It can be observed that the blue fluid's radius expanded far faster than the yellow's because of its lower viscosity. Figures 13-18 show the logarithmic r-t relationship, where a straight line is observed for every fluid.

(4) While it is clear that the trendline from the collected data follows the trendline of the theoretical data, it is also evident that there is a sizable difference between the expected and experimental diameters for both the yellow and blue fluids. In order to close this gap, we ascertained the experimental viscosities by altering the theoretical data using Huppert's equation;

$$R(t) = 0.779 \left(\frac{g\rho V^3 t}{\mu} \right)^{1/8}$$

This trial-and-error process was repeated with different values of μ until the transformed theoretical data began to resemble the measured values from the lab. Once the comparison between measurements and predicted values was reasonable, we recorded the value of μ. These are displayed in the table below.

Calculated Experimental Viscosity					
Blue, 30 mL	Blue, 40 mL	Blue, 50 mL	Yellow, 30 mL	Yellow, 40 mL	Yellow, 50 mL
8.95	7.16	5.37	415.1	59.3	59.3

Table I: Experimental viscosities as obtained from the data. These values were calculated from the transformed theoretical values of diameter as predicted by Huppert's equation.

The full experimental diameter data, theoretical diameter data, and transformed theoretical diameter data are displayed in Appendix B below. Evidently, these experimental values are significantly different from the known values of 0.197 ($Pa \cdot s$) for the blue fluid and 5.93 ($Pa \cdot s$) for the yellow liquid. The percent error calculations are shown below.

% Error -- Calculated Experimental Viscosity					
Blue, 30 mL	Blue, 40 mL	Blue, 50 mL	Yellow, 30 mL	Yellow, 40 mL	Yellow, 50 mL
4443.1	3534.5	2625.9	6900	900	900

Table II: Percent error calculations for the calculated viscosity values of the blue and yellow fluids in each trial.

There are several factors that may have impacted the accuracy of the previously discussed results. For this experiment to be improved, we would like to reduce errors in both the procedural setup as well as the execution of each trial in the future. For instance, the setup could be properly calibrated on a completely level surface before the trials are conducted. In our experiment, we noted the fluid began to distort in a certain direction due to the uneven plane it was poured on. Likewise, using the plastic wrap as a pour surface was a practical method to rapidly repeat trials, but likely hindered the accuracy of our results due to its bunching and inability to be completely flattened. To produce more accurate results, using a flat plastic reusable surface would be preferable but would likely require more time in between trials. Additionally, the process of pouring was uncontrolled and could be more precise through a pouring apparatus which controls the rate and location of pour for each trial. These improvements would enhance the precision and accuracy of the experiment.

(5) To calculate the Reynolds and Bond numbers of each fluid, we first had to compute the mean thickness H, using $H = \frac{V}{\pi R^2} = \frac{4V}{\pi D(t)^2}$, using D(t) from our diameter-time graphs. Our Reynolds and Bond numbers could then be calculated using:

$$Re(t) = \frac{\rho\, U(t)\, H(t)}{\mu} \qquad Bo(t) = \frac{\rho g\, H(t)^2}{\sigma}$$

The following table presents our calculations of Re and Bo:

	Run	t [s]	D [mm]	R [m]	H [mm]	Re	Bo
1	Yellow, 30mL	1.0	80	0.04	5.968	0.00499	6.3377
2	Yellow, 30mL	3.0	95	0.0475	4.23	0.00295	3.1871
3	Yellow, 30mL	5.0	105	0.0525	3.46	0.00164	2.1357
4	Yellow, 40mL	1.0	90	0.045	6.29	0.00701	7.0340
5	Yellow, 40mL	3.0	110	0.055	4.21	0.00352	3.1521
6	Yellow, 40mL	5.0	120	0.06	3.54	0.00148	2.2256
7	Yellow, 50mL	1.0	9	0.0475	7.05	0.00787	8.8532
8	Yellow, 50mL	3.0	115	0.0575	4.81	0.00403	4.1229
9	Yellow, 50mL	5.0	125	0.0625	4.07	0.00170	2.9536
10	Blue, 30mL	1.0	100	0.05	3.82	0.13049	2.6646
11	Blue, 30mL	3.0	118	0.059	2.74	0.07810	1.3743
12	Blue, 30mL	5.0	130	0.065	2.26	0.03646	0.9329
13	Blue, 40mL	1.0	110	0.055	4.21	0.15977	3.2354
14	Blue, 40mL	3.0	130	0.065	3.01	0.10009	1.6585
15	Blue, 40mL	5.0	145	0.725	2.42	0.04598	1.0716
16	Blue, 50mL	1.0	120	0.06	4.42	0.1678	3.5694
17	Blue, 50mL	3.0	140	0.07	3.25	0.10789	1.9267
18	Blue, 50mL	5.0	155	0.0775	2.65	0.05029	1.2824

As shown in the table above of our calculations of the Reynolds and Bond numbers of both fluids at three different volumes, most of our experimental calculations of the Reynolds and Bond numbers satisfy the theoretical conditions needed. Our calculated Re is orders of

magnitude below 1 for all volumes, and our calculated Bo is above 1 for every calculation other than the blue fluid at 30mL, which is the result of experimental error.

(6) Figures 19-24 show the R* vs T* plots, where $R^* = \frac{R}{V^{1/3}}$ and $T^* = \frac{t\rho g V^{1/3}}{\mu}$.

[R = radius, V = volume, t = time, ρ *= density, and* μ *= viscosity]*

The idea behind these equations is to remove the influence of the individual fluid's properties (viscosity, density), as well as that of the different volumes, giving a single consistent graph across all six measurements. The six graphs are almost identical, showing that there is a consistent relationship between radius and time for any gravity current whereby the radius expands at an exponentially decreasing rate.

Conclusion

Ultimately, this lab demonstrated the spreading behavior of viscous gravity currents and compared experimental results with the theoretical predictions from Huppert's equation. While our data showed the expected trends—greater volume led to faster spreading, and the lower-viscosity blue fluid advanced more quickly than the yellow fluid—numerous quantitative discrepancies arose due to errors in measurement, setup, and data collection. Despite these challenges, our nondimensional analysis and Reynolds/Bond number calculations supported the theoretical assumptions of radial expansion. Overall, the experiment confirmed the laws viscous gravity currents held true while highlighting the need for improved setup and measurement precision in future trials.

Appendix A: Error & Uncertainty Calculations

Error Table					
Blue, 30 mL	Blue, 40 mL	Blue, 50 mL	Yellow, 30 mL	Yellow, 40 mL	Yellow, 50 mL
64.23587339	43.41652517	57.58064617	48.85538371	20.66186488	12.23154396
47.71875106	41.36147049	46.3554859	47.10855654	18.87252701	14.17940527
42.90557758	41.75461993	41.29766818	45.62411414	20.86409523	15.02298075
41.24176941	33.89945214	37.49347721	45.41515349	21.22364585	18.03677813
38.75507098	32.51088717	33.73019751	45.13474153	22.07218431	19.48254072
38.12642626	31.06344362	33.57050661	43.91828188	21.97680329	19.75404927
37.97349759	32.66335217	33.12048332	44.05319801	22.28775174	21.2817838
36.34982697	32.54039058	32.72651606	44.80025616	23.73348277	20.53823325
36.6340936	32.95006862	33.62777771	44.00806697	24.77828429	21.13089398
38.76810118	33.47744889	33.24985298	43.07453081	23.07696716	22.1798718
38.7818298	34.37669832	33.73185035	43.42884429	23.88949066	23.10150215
39.3190171	34.48403948	33.77232315	42.47870974	23.1669631	22.80649909
37.14310083	34.29459106	34.19094122	43.43991291	22.27738776	23.54446337
37.97844677	32.90223391	34.74531678	42.82194035	23.51861841	23.72100801
35.5483103	33.91433826	35.02846766	42.00666845	26.32367889	24.36609321

Total Weighted Error					
Blue, 30 mL	Blue, 40 mL	Blue, 50 mL	Yellow, 30 mL	Yellow, 40 mL	Yellow, 50 mL
40.76531285	35.04063732	36.94810072	44.41122393	22.58158302	20.09184312

The error presented in these two tables is representative of the error between our experimental values for the diameter of the fluid and the theoretical (predicted) values of the fluid as described by Huppert's prediction. The data in blue represents the error in each individual point, and the data in orange represents the overall error in our experimental values for each of the six trials that were conducted. The error was calculated using the standard formula:

$$\% \text{ error} = \left| \frac{\#\,\text{experimental} - \#\,\text{actual}}{\#\,\text{actual}} \right| \times 100$$

Hence it is possible to observe a large gap between the experimental values and theoretical values predicted by Huppert's equation.

Appendix B: Experimental and Predicted Data

Blue, 30 mL				
Time_s	Diameter_mm	Radius_mm	d_theoretical_mm	adjusted mu
0.56666667	42.4723152	21.2361576	118.7567521	72.82610134
1.13333333	67.7069132	33.8534566	129.5051564	79.41742656
1.7	77.7843366	38.8921683	136.2380655	83.54629933
2.26666667	82.9821188	41.4910594	141.2263745	86.60531782
2.83333333	88.9405268	44.4702634	145.2210464	89.05500067
3.4	91.9247418	45.9623709	148.5686639	91.10788545
3.96666667	93.9448244	46.9724122	151.4591678	92.88045106
4.53333333	98.0266471	49.01332355	154.0084534	94.44376875
5.1	99.0362703	49.51813515	156.2926752	95.84453936
5.66666667	96.969697	48.4848485	158.3646741	97.11516689
6.23333333	98.1098793	49.05493965	160.2626785	98.27909449
6.8	98.3124624	49.1562312	162.0152768	99.35385359
7.36666667	102.861815	51.4309075	163.644431	100.3529122
7.93333333	102.439383	51.2196915	165.1673937	101.2868501
8.5	107.37521	53.687605	166.5979752	102.1641364

Blue, 40 mL				
Time_s	Diameter_mm	Radius_mm	diameter_theoretical_mn	adjusted_mu
0.36666667	70.8872164	35.4436082	125.2790088	78.99885734
0.73333333	80.1106265	40.05531325	136.6177276	86.14886466
1.1	83.7105024	41.8552512	143.7204158	90.62770158
1.46666667	98.4783733	49.23918665	148.9826885	93.94600315
1.83333333	103.391129	51.6955645	153.1967523	96.60332163
2.2	108.043041	54.0215205	156.7282247	98.83020937
2.56666667	107.588798	53.794399	159.7774785	100.753018
2.93333333	109.599451	54.7997255	162.4667737	102.4488428
3.3	110.549545	55.2747725	164.8764476	103.9683431
3.66666667	111.134066	55.567033	167.062243	105.3466693
4.03333333	110.945699	55.4728495	169.064488	106.6092516
4.4	111.975517	55.9877585	170.913341	107.7751075
4.76666667	113.428542	56.714271	172.6319702	108.8588465
5.13333333	116.910192	58.455096	174.2385758	109.8719452
5.5	116.144048	58.072024	175.7477264	110.8235905
5.86666667	115.390502	57.695251	177.1712731	111.7212553
6.23333333	118.095453	59.0477265	178.5189909	112.5711037
6.6	117.658342	58.829171	179.7990411	113.3782821
6.96666667	120.308076	60.154038	181.0183109	114.1471334
7.33333333	118.095453	59.0477265	182.1826678	114.8813574

Blue, 50 mL				
Time_s	Diameter_mm	Radius_mm	diameter_theoretical	adjusted_mu
0.5	60.0649242	30.0324621	141.5979235	92.55858951
1	82.8344418	41.4172209	154.4136305	100.9358576
1.5	95.3569594	47.6784797	162.4415188	106.1834629
2	105.25427	52.627135	168.3892581	110.0713332
2.5	114.747653	57.3738265	173.1522484	113.1847663
3	117.675684	58.837842	177.1437324	115.7938874
3.5	120.777842	60.388921	180.5901836	118.0467359
4	123.534156	61.767078	183.6297881	120.03364
4.5	123.686858	61.843429	186.3533474	121.8139543
5	126.040208	63.020104	188.8238658	123.4288628
5.5	126.629769	63.3148845	191.0869244	124.9081607
6	127.936381	63.9681905	193.17661	126.2741296
6.5	128.406049	64.2030245	195.1191088	127.5438866
7	128.509305	64.2546525	196.9349917	128.7308783
7.5	129.059923	64.5299615	198.6407252	129.8458684
8	130.400872	65.200436	200.2497039	130.8976126
8.5	128.298734	64.149367	201.7729761	131.8933329
9	128.866088	64.433044	203.2197664	132.8390591
9.5	130.16201	65.081005	204.597859	133.7398796
10	130.830504	65.415252	205.9138857	134.6001294
10.5	133.030725	66.5153625	207.1735445	135.423533

Yellow, 30 mL				
Time_s	Diameter_mm	Radius_mm	d_theoretical	adjusted_mu
1.4	77.9350371	38.96751855	152.3817026	89.59743293
2.8	81.4745246	40.7372623	154.0410306	90.57308501
4.2	84.2935194	42.1467597	155.0200389	91.14872256
5.6	84.9986646	42.4993323	155.7184274	91.55936124
7	85.7337179	42.86685895	156.2623057	91.87915093
8.4	87.8845929	43.94229645	156.7080964	92.14126707
9.8	87.8845929	43.94229645	157.0859991	92.36346642
11.2	86.8921744	43.4460872	157.4140899	92.55637732
12.6	88.301549	44.1507745	157.7040552	92.7268712
14	89.9216859	44.96084295	157.9638906	92.87964928
15.4	89.4951775	44.74758875	158.1993091	93.0180707
16.8	91.1220849	45.56104245	158.4145357	93.14461969
18.2	89.7115285	44.85576425	158.6127835	93.26118549
19.6	90.7967881	45.39839405	158.7965535	93.36923862
21	92.1907411	46.09537055	158.9678307	93.46994621

Yellow, 40 mL				
Time_s	Diameter_mm	Radius_mm	d_theoretical	adjusted_mu
1.33333333	75.1626472	37.5813236	94.73709848	71.04280156
2.66666667	83.8140405	41.90702025	103.3115385	77.47272448
4	86.0070018	43.0035009	108.6826543	81.50049311
5.33333333	88.751041	44.3755205	112.6620316	84.4846051
6.66666667	90.2783958	45.1391979	115.8487441	86.87430238
8	92.472527	46.2362635	118.5192749	88.87691795
9.33333333	93.8959387	46.94796935	120.8251477	90.60607857
10.6666667	93.7001406	46.8500703	122.8588167	92.1311152
12	93.7872125	46.89360625	124.6810334	93.49758497
13.3333333	97.1799068	48.5899534	126.3339512	94.73709848
14.6666667	97.3058148	48.6529074	127.8480668	95.87252495
16	99.3037696	49.6518848	129.2461858	96.92096628
17.3333333	101.463628	50.731814	130.5458284	97.89556077
18.6666667	100.772448	50.386224	131.7607579	98.80662933
20	97.9172976	48.9586488	132.9019909	99.66243341
21.3333333	98.6077969	49.30389845	133.9784895	100.4696935
22.6666667	98.1670605	49.08353025	134.9976457	101.2339528

Yellow, 50 mL				
Time_s	Diameter_mm	Radius_mm	d_theoretical	adjusted_mu
1.4	90.9595819	45.47979095	103.635846	77.71592076
2.8	96.9907385	48.49536925	113.0156914	84.74981254
4.2	101.030301	50.5151505	118.8913213	89.15591339
5.6	101.015151	50.5075755	123.2444854	92.42032592
7	102.040202	51.020101	126.7305289	95.03448976
8.4	104.040404	52.020202	129.6519052	97.22521296
9.8	104.045307	52.0226535	132.1743708	99.11679532
11.2	106.795871	53.3979355	134.3990643	100.7850801
12.6	107.571501	53.7857505	136.3924437	102.2799037
14	107.547901	53.7739505	138.2006217	103.635846
15.4	107.547901	53.7739505	139.8569595	104.8779241
16.8	109.141116	54.570558	141.3864052	106.0248465
18.2	109.184718	54.592359	142.8081246	107.0909857
19.6	109.946383	54.9731915	144.1371735	108.0876317
21	109.960812	54.980406	145.3856037	109.0238223